Convertible Bond Trading

Cindy Pan
Susquehanna University
pan@susqu.edu

March 31, 2026

Abstract

We investigate how convertible bonds trade when the conversion option is in-the-money (ITM). Empirical results show trading activity in convertible bonds increase when the option is ITM. ITM convertibles experienced an average increase in trading volume of around 9.7 million and an average increase of 4 trades on event day relative to the estimation window. We examine whether ITM convertible bonds lead to a spillover effect on a firm's other bonds and find an increase in trading activity in those other bonds. We explore the equity market's reactions to when convertible bonds are convertible and find increases in trading activity (lit and hidden liquidity). On average, abnormal lit volume increased by 1.29 million and abnormal hidden volume increased by 234,000 on event day.

JEL Classifications: G1, G12, G14, G30

Keywords: Convertible Bonds, Corporate Bond Market, Bond Trading, Equity Markets

1. Introduction

Convertible bonds are hybrid securities incorporating features of debt, equity, and options. Investors enjoy the limited downside risk of convertible bonds while holding the option to convert the bond into the issuing company's stock and benefit from the equity upside. A sizable portion of the corporate bond market consists of bonds that are convertible. In 2022, convertible bond issuance in the United States is $8.5 billion with an average daily trading volume of $2 billion.[1] Globally, convertible bond issuance totals $74.1 billion in 2022.[2] While convertible bonds are a substantial size of the fixed income market, it remains largely unknown what happens to these bonds when the conversion option is at-the-money (ATM) or gets in-the-money (ITM) [3].

Bessembinder, Spatt, and Venkataraman (2020) survey the literature that studies fixed-income trading, and there was no mention of convertible bonds as corporate bond research often filters out convertibles due to the option-like conversion feature. Research on convertible bonds often center around convertible bond calls (Cowan, Nayar, and Singh, 1993; Ingersoll, 1977b), convertible arbitrage (Ammann, Kind, and Seiz, 2010; Choi, Getmansky, and Tookes, 2009), or pricing of convertible bonds (Ammann, Kind, and Wilde, 2003; Batten, Khaw, and Young, 2018). In the Chinese market, Chen, Xu, and Wang (2023) find that convertible bond trading contains information and positively predicts future stock returns.

[1] See https://www.sifma.org/resources/research/us-fixed-income-securities-statistics/
[2] See https://dealogic.com/insight/ecm-highlights-fy22/
[3] The conversion option is in-the-money (ITM) when the conversion value of the common stock to be received in the conversion exchange equals or exceeds the conversion price (stock price > conversion price). Trading in the options market concentrates on at-the-money (ATM) options (Blasco, Corredor, and Santamaria, 2010; Heng and Leung, 2023; Mohil, Nayyar, and Patro, 2020). We will use ITM to refer to ATM and ITM convertibles as both statuses mean the option to convert is beneficial to the bondholder.

With very little that is known about the trading patterns of convertible bonds in the U.S. market, this paper seeks to explore how they trade when the option to convert is ITM. In a market that experiences sparse trading and a unique security with features of debt, equity, and options, it is interesting to examine changes in trading behavior when a particular feature suddenly outweighs the others. Easley, O'Hara, and Srinivas (1998) find that changes in stock prices lead option volumes as a and that option volumes lead to stock price changes as option markets serve as a venue for information-based trading. Pan and Poteshman (2006) find that option trading volume contains information about future stock prices.

In convertibles, stock price increases that lead the convertibles to ITM status may result in increased convertible bond trading volume. Alternatively, information-induced trading in convertible bonds may lead to an increase in stock prices, which drive convertibles to be ITM. We are not trying to disentangle the direction of the effect as our interest lies in when the conversion feature is ITM, which is only possible when the stock price exceeds the conversion price. Our empirical results show trading activity in convertibles do increase significantly when the option is ITM.

Additionally, we examine spillover effects to the non-convertible bonds of the firms' with ITM convertibles and how those firms' equity trade during the event window when the convertibles are in-the-money. Even though post-trade transparency in the bond market has increased after public dissemination through the Trade Reporting and Compliance Engine (TRACE), pre-trade transparency remains nonexistent (Bessembinder, Spatt, and Venkataraman, 2020).[4] ITM convertibles provide traders with a glimpse of the demand for these bonds based on

[4] Quotations are distributed only to some market participants (Bessembinder, Spatt, and Venkataraman, 2020).

the equity performance of the firms. Trading activity in convertibles may also serve as a proxy for information efficiency as traders in this illiquid market reevaluate future prospects of the underlying firms. We find that the increase in equity prices does lead to a spillover effect of increased trading in the firms' non-convertible bonds. While it is more intuitive that equity trading on lit exchanges increases when there are informational events that increase equity prices, we find that hidden liquidity in those firms also increases.[5]

The remainder of the paper proceeds as follows. Section 2 provides a detailed review of the literature on the corporate bond market focusing on convertible bonds and outline the development of the hypotheses. Section 3 describes the data and methodology. Section 4 presents the results of the analyses, and Section 5 concludes.

2. Literature Review and Hypotheses Development

The over-the-counter (OTC) corporate bond market is often characterized as illiquid and costlier to transact in relative to the equity market (Bao, Pan, and Wang, 2011; Bessembinder, Jacobsen, Maxwell, and Venkataraman, 2018; and Goldstein and Hotchkiss, 2020). As a convertible bond becomes ATM or ITM, the bond price mimics the underlying stock price. In other words, the convertible bond becomes linked to the firm's equity with bond trading patterns following stock trading patterns. We hypothesize that when the option to convert the bond to equity has a greater payoff than holding the bond until maturity, demand for the convertible bond should increase as investors try to capture the benefit of the conversion option. When ITM convertible bonds are more actively traded, dealers are less exposed to inventory risk and

[5] Equity exchanges are also referred to as lit exchanges where price quotes are publicly displayed. Hidden liquidity refers to hidden limit order trading on lit exchanges when traders place hidden limit orders where price quotes are not publicly displayed (Bloomfield, O'Hara, and Saar, 2015). Bartlett and O'Hara (2024) find that hidden orders provide liquidity for approximately 40% of trading volume.

experience reduced search costs which might result in a more liquid market (Goldstein and Hotchkiss, 2020).[6]

H1: When the convertible option is in-the-money, demand for convertible bonds will increase, resulting in higher trading activity.

As convertible bonds approach ITM status or are ITM, this should indicate that the stock is performing well in that the future performance of the firm is on an upward trend. If the ITM status of convertible bonds relates to future performance, we anticipate there will be spillover effects to the firm's other bonds. Demand for the convertible bonds should also increase demand for the firm's other bonds as investors anticipate better future performance.

H2: In-the-money convertible bonds lead to increased trading activity in the firm's other bonds.

The extant literature has mixed results of whether a lead-lag relationship exists between the stock and bond market. Tolikas (2018) finds that stock returns only lead the returns of high-yield bonds and that the stock market is more efficient relative to the bond market by incorporating information at a faster rate. On the other hand, Ronen and Zhou (2013) find that the majority of price discovery for the bonds with the highest institutional trade volume immediately after an earnings announcement occurs before the equity market opens.

Downing, Underwood, and Xing (2009) find that stock returns lead bond returns for convertible bonds in all rating classes but only those rated BBB and high-yield grade for

[6] In the bond market, large trades may reflect investors waiting for relevant and material information and then quickly acting on it in large trades (Ronen and Zhou, 2013). In the equity market, large trades are exposed to adverse selection and information leakage so using large trade sizes as identification for institutional investors in the bond market is not subject to the same scrutiny as in the equity markets.

nonconvertible bonds. The return predictability of convertible bonds depends on the extent to which the option is ITM. As the credit quality of the convertible bonds improve, they become more equity-like because the conversion option goes more into the money when firms' prospects improves (Downing, Underwood, and Xing, 2009). The findings of Batten, Khaw, and Young (2018) suggest that equity-like convertible bonds are more attractive to investors, such as hedge funds, because of the higher value of the conversion option.[7]

In the case of convertible bonds, the equity performance determines the value of the option. If the stock price continues on an upward trend leading the convertible bond deeper ITM, this may indicate that investors are adjusting their valuation of the stock upwards until its new intrinsic value is realized. Downing, Underwood, and Xing (2009) find that stock returns predict returns of convertible bonds because the convertible option is highly sensitive to firm-specific information.

Institutional investors typically hold both bonds and stocks in their portfolios, so we expect that they are actively trading in both debt and equity markets (Holden, Mao, Nam, 2018; Chang and Yu, 2010). In this situation, equity volume and trade size should increase due to more trading activity. On the other hand, ITM convertibles may lead to a downward demand curve for stocks (Harris and Gurel, 1986). While it is possible that demand will shift from stocks to the convertibles, another reason why the equity market might experience increased trading might be due to high-frequency traders trading along with informed institutional investors (Van Kervel and Menkveld, 2019).

[7] Hedge funds are one of the most common acquirers of convertible bonds purchasing 70% to 80% of convertible debt offerings (Choi, Getmansky, and Tookes, 2009).

If informed traders are incorporating new information into their trades, they might prefer to engage in hidden limit order trading, rather than lit trading, to avoid information leakage.[8] In this case, hidden liquidity in the stocks of firms with convertible bonds that are in-the-money should increase. If traders, informed or liquidity, simply prefer hidden orders, then hidden liquidity will increase as equity trading increases.

H3: Demand for firms with in-the-money convertible bonds will lead to increased equity trading.

H4: Hidden liquidity in firms with convertible bonds that are approaching or are in-the-money will increase.

3. Data and Methods

Data Sources

Bond transaction data are obtained from enhanced TRACE from January 1, 2015 to December 31, 2022. Bond characteristics are sourced from the TRACE master file such as the indicators for 144A, convertible features, and investment or noninvestment grade, bond maturity date, and coupon rate. Information regarding features of convertible bonds like conversion price, conversion start date, issue amount, and type of convertible bond are obtained from Bloomberg Financial Database. Information on firm characteristics and financials are obtained from CRSP and Compustat.

Sample Selection

[8] MIDAS only reports the executed portion of hidden orders. While data limitations do not allow us to analyze all submitted nondisplayed orders, the portion of hidden orders that are executed is our main focus as these orders are successfully traded. This limitation also prevents us from examining other aspects of the order exposure decision such as price aggressiveness and order size.

The sample only examines convertible bonds designated with the convertible indicator on the TRACE master file. The sample excludes 144A bonds, issues denominated in foreign currency, bonds with callable and putable conversion features, bonds of financial and utility firms (SIC 6000-6999 and 4900-4999), and firms that are not designated with share code 10 or 11. We follow Dick-Nielsen (2009) to remove duplicate, canceled, corrected, reversed, and commission trades from TRACE. A bond must have at least 10 trades in the sample period with maturity between one and fifty years to be included for analysis.

The convertible bonds must stay ITM for at least five days and trade every day in an estimation window of [-30, -6]. This ensures that the convertible bonds are relatively more liquid and have enough trading activity prior to being ITM to compare with its trading activity after becoming ITM. The first time a convertible bond stays ITM for at least five days is used as the event where the first of the five days is used as event day. Figure 1 provides a visual timeline of the convertible status of two hypothetical convertible bonds, CV Bond A and CV Bond B.

While CV Bond A went ITM for the first time on April 5th, it fell out-of-the-money (OTM) on April 8th, which is less than the minimum required five days, so April 5th is not used as the event day for CV Bond A. However, on May 15th, CV Bond A went ITM a second time and fell OTM on May 25th, which is longer than the minimum five days, so May 15th is used as the event day for CV Bond A. While CV Bond A went ITM a third time on June 10th and stayed ITM until August 11th, this period is not used as the event window, since we are interested in the trading patterns of convertible bonds when they first get ITM with a long enough event window. Similarly, the first time CV Bond B goes ITM for at least five days, the event day is the first day of the period that it is ITM.

Table 1 reports the sample selection process. The convertible bonds are identified based on the convertible flag indicator in the TRACE Master File, which contains matured and active convertible bonds, and merged with enhanced TRACE. After following the standard TRACE filters, there are 4,163 convertible bonds identified in TRACE. To identify when the convertible bonds are ITM, we need its conversion price, which is not provided in TRACE. We identified convertible bonds in Bloomberg with a conversion price to merge with the transaction data in TRACE. The Bloomberg sample only includes active convertible bonds in the sample period (not matured or defaulted). The convertible bond sample from Bloomberg consists of 582 convertible bonds after excluding convertible bonds denominated in foreign currency and with callable and putable conversion features.[9] Merging TRACE and Bloomberg resulted in a sample size of 298 convertible bonds. After excluding 7 bonds with less than ten trades in the sample period, 148 bonds that did not stay ITM for at least five days over the entire sample period, and 96 bonds that did not have the full [-30, -6] estimation window, the final sample includes 47 convertible bonds from 41 issuing firms.

Table 2 reports the descriptive statistics over a [-5, +5] event window for the convertible bonds that went ITM. The average number of trades for ITM convertibles over the 11-day window is 4.93 trades. These bonds have, on average, 4.80 years to maturity and an average issue amount of $412,347,000. The average stock price for these firms with ITM convertible bonds is $63.91.

While it is important to limit the convertible bond sample to the bonds that stay ITM for at least 5 days for the analyses, it is informative to examine how often the convertible bonds

[9] Without access to Mergent FISD, which is the standard dataset for bond characteristics, we rely on Bloomberg for bond characteristic variables.

become convertible. Since the option to convert is dependent on the stock price movement relative to the conversion price, the convertible bond may fall in- and out-of-the-money multiple times. Table 3 reports the number of times the sample convertible bonds went ITM. The greatest number of times a convertible bond became ITM is 23 times over the sample period which suggests volatile stock prices. Five bonds went ITM once and stayed in the money over the entire sample period which implies the increase in stock price lead these bonds deep ITM or the stock price of these firms were not volatile. Eight bonds experience fluctuations in the convertible option seven times over the sample period.

We suspect that earnings announcements might be pushing stock prices to new highs causing convertible bonds to go ITM. Therefore, we expect most of the convertible bonds to go ITM around earnings season in January, April, July, or October. Panel A table 4 reports the number of convertible bonds that went ITM during a particular month over the sample period. July and November appear to have the greatest number of convertible bonds go ITM. Although July having one of the most convertible bonds go ITM aligns with our expectations of earnings announcements pushing stock prices past conversion prices, November having the greatest number of ITM convertible bonds is surprising. Panel B table 4 reports the number of convertible bonds that went ITM each year in the sample period. 2020 and 2021 have the greatest number of convertibles go ITM which may be a result of volatile equity prices in 2020 during the COVID-19 pandemic and the rallying of equity prices post-pandemic in 2021.[10]

[10] There may be concern that the increases in the Federal Funds Rate starting in the year 2022 may influence demand for convertible bonds as yields increase, but panel B table 4 shows that only 7 convertibles that went in-the-money in 2022 is included in our sample. The small number of convertibles in 2022 is unlikely to influence our results.

Taking a closer look at what kind of information event that may have caused a firm's stock price to increase, Table 5 reports the types of information event that may have cause the convertible bonds to go ITM. Earnings Announcements include news related to quarterly earnings announcements, earnings estimates from the firm or analysts, or news regarding when the firm is set to announce earnings. Macroeconomic news include news related to how the overall economy is related to the firm's operations. Firm-specific news include news related to mergers and acquisitions, partnerships, patent issuances, government contracts, CEO turnover, new product releases, etc. Analyst forecasts include news regarding analyst upgrades of stock price targets, earnings targets, EPS targets, or future forecasts. Unknown include firms with no tangible news that might affect stock prices. Firm-specific news and analyst forecasts appear to be the dominant types of information event that occurs on the day a convertible bond goes ITM. In unreported analyses, combining the information provided in Table 4 and Table 5 show that the main type of event that resulted in a convertible bond to go ITM is firm-specific announcements for July. In November, the two dominant events that resulted in a convertible bond to go ITM is firm-specific announcements and analyst forecasts.

Methods

We first start with an 11-day event study around the event of when the convertible bond goes ITM (stock price >= conversion price). We compute abnormal trading measures of abnormal quantity traded, abnormal number of trades, and abnormal trading volume using the following equation:

$$Abnormal\ Trading\ Measure = Trading\ Measure_{i,t} - \overline{Trading\ Measure_{i}}, \quad (1)$$

where $Trading\ Measure_{i,t}$ is either the daily average quantity, the daily average number of trades, or daily average volume for firm *i* on day *t* and the $\overline{Trading\ Measure_{i}}$ is the average of either the daily average quantity, the daily average number of trades, or the daily average volume for firm *i* measured in a 25-day estimation window [-30, -6]. $Abnormal\ Trading\ Measure$ is either abnormal quantity, abnormal number of trades, or abnormal volume.

Next, we run OLS regressions using the following equation:

$$Y_i = \beta_0 + \gamma X_i + \varepsilon_i\ , \qquad (2)$$

where Y_i are the abnormal bond trading measures of firm i. X_i is the set of control variables including years to maturity, issue amount, buy/sell side indicator (representing the side of a trade where side is +1 for buys, -1 for sells, and 0 for no trade), firm size, sales, ROA, leverage and dummy variables that capture the seven trading days around the event date (i.e., Event_{t-3} is a dummy coded 1 for three days before day 0 and 0 otherwise, while Event_{t+3} is a dummy coded 1 for three days after day 0 and 0 otherwise). Firm size, sales, and bond issue amount are in logs and the buy/sell side variable is log(1+side) transformed.

4. Empirical Findings

Trading of In-the-Money Convertibles

Figure 2 provides a graphical representation of the abnormal trading metrics related to when the convertible bonds become ITM on day 0 over the window of [-5, +5]. Panel A shows that abnormal quantity peaks at day -1 and remains elevated over the event window. Panel B (panel C) shows that the abnormal number of trades (abnormal volume) peaks at day 0 and falls back to normal levels over the event window.

The 11-day event study around ITM convertible bonds is reported in Table 6. Column (1) reports the results for abnormal trading quantity. Column (2) reports the results for abnormal number of trades and Column (3) reports the results for abnormal trading volume. The t-test in Table 6 shows that only the abnormal number of trades and abnormal volume are positively significant on day 0. The 4.23 coefficient on abnormal trades on day 0 indicates that the first day the convertible bond went ITM, there were around 4 more trades than the previous 25 days. The 9,693,890.21 coefficient on abnormal volume on day 0 indicates that the first day the convertible bond went ITM, there was around 9.7 million more trade volume than the previous 25 days. Overall, trading interest in ITM convertible bonds increases on the day the bonds go ITM.

To control for some firm and bond characteristics, Table 7 reports the OLS regression results where seven dummy variables are included to capture the seven days around the event day. Regression results in Table 7 report similar findings as the t-test. Abnormal trades and abnormal volume are positively significant on event day after controlling for firm and bond characteristics. Firm sales is positively related to abnormal trades and abnormal volume, while ROA is negatively related to abnormal trades and abnormal volume. Overall, trading activity for convertible bonds increase when the convertible option is ITM, supporting hypothesis 1 which states that demand for ITM convertible bonds will increase resulting in higher trading activity.

Table 8 reports the direction of trades and the initiating party of the trades pre- and post-event. The event window pre-event is [-5, -1] and the post-event window is [0, +5]. The t-test shows that buys and sells significantly increased in the post-event window, which confirms the increase in trading activity after the convertible bonds go ITM. More trades are customer-initiated but dealer trades also increased in the post-event window.

Spillover Effects in Non-Convertible Bonds

To see if demand for other bonds of the firms with ITM convertible bonds change after the convertible bonds go ITM, we test for spillover effects using a sample of the firms' non-convertible bonds. After using the transaction filtering process as the convertible bonds, Table 9 reports the descriptive statistics for the sample of non-convertible bonds issued by the same firms with ITM convertible bonds during the sample period. This sample consists of 80 unique non-convertible bonds from 18 issuing firms. The average number of trades in the non-convertible bonds over the [-5, +5] event window of when the firm's convertible bonds went ITM is 1.33 trades. The average volume in the non-convertible bonds is 802,521 with a minimum of zero trading volume and a maximum of over 8,000,000. The average stock price for this sample of firms is $54.41.

Figure 3 presents some graphical evidence of increased trading activity of the firms' non-convertible bonds when their convertible bonds go ITM. Panel A shows that abnormal quantity starts on an upward trend from day -3 to day 0. Then, there is a sharp decrease on day +1, but abnormal quantity starts to dramatically increase on day +2 before leveling off through day +5. Panel B shows that abnormal trades spikes around day -1 and day 0 before a steep decline on day +1 before increasing again on day +2. Panel C shows that abnormal volume reaches a peak on day 0 before falling around average levels and spiking once more on day +2. Overall, the trading metrics over the event window indicates an increase in trading activity in non-convertible bonds of firms when their convertible bonds go ITM.

To test for spillover effects related to hypothesis 2, Table 10 reports the t-test during an 11-day event study where day 0 is the day the firms' convertible bonds become ITM and tests the significance of changes in trading activity related to the firms' non-convertible bonds. Column 1 reports a negatively significant effect on abnormal quantity on day -4 which indicates a decrease in trading activity related to non-convertible bonds in the four days before convertible bonds go ITM. Column 2 reports a significant increase in abnormal trades on days -1, 0, and +2 which indicate increased trading activity in the non-convertible bonds of firms with ITM convertible bonds. However, the negative and significant coefficient on day +1 indicates that trading in non-convertible bonds of firms with convertible bonds decreased the day after the convertibles go ITM. Column 3 reports a significant decrease in abnormal volume on day -4 and +1. This indicates a decrease in trading volume in non-convertible bonds four days prior and one day after the firms' convertible bonds go ITM. The mixed results on the trading metrics over the event windows warrants a closer look at whether trading activity changed for the non-convertible bonds of firms with convertible bonds that went ITM.

To further test for spillover effects in non-convertible bonds, regression results controlling for firm and bond characteristics are reported in Table 11. After controlling for the independent variables, all three abnormal trading metrics have a positive and significant coefficient on event day. On event day, abnormal quantity is 166,319 which means quantity traded in non-convertible bonds increased by 166,319 compared to the estimation window. The 1.27 coefficient on abnormal trades on event day indicates that the first day the convertible bond went ITM, there was around 1 more trade in the non-convertible bonds than in the previous 25 days. The coefficient of 2,225,700 on abnormal volume indicates an increase in volume on event day relative to the estimation window. Abnormal trades was 1.31 trades higher the day prior to the event compared to the estimation window. ROA is negatively related to all trading metrics. Overall, regression results indicate that trading activity increased in the firms' non-convertible bonds. These findings provide support to hypothesis 2 in that trading activity increased in non-convertible bonds of firms with ITM convertible bonds.

Equity Market Trading Activity

To test trading activity in the equity market, we examine the firms with convertible bonds around the [-5, +5] when the convertible bonds go ITM. We are interested in the trading activity in the firms' stock, so we examine total trade volume, total trades, and canceled trades. Table 12 provides descriptive statistics related to the equity market for the firms with convertible bonds that went ITM. The average stock price for the firms is $63.92 with average sales of $2.29 million. The sample firms have an ROA of -7% and leverage of 50%. The average total volume is 1.96 million and the average total trade is 12,310. The average canceled trades is 142,530.

Figure 4 presents graphical evidence of increased trading activity in the stocks of firms with ITM convertible bonds. Panel A shows that abnormal lit volume starts to increase on day t-1 and peaks on the day convertible bonds go ITM. Abnormal lit volume declines to normal levels on day +4 and day +5. Panel B shows that abnormal hidden volume peaks on day 0 and declines to normal levels over the event window.[11] Panel C shows that abnormal canceled trades are above normal levels throughout the event window and peaks on day 0. Figure 4 provides initial evidence of increased trading activity in the equity market related to firms with convertible bonds that went ITM.

To formally tests the graphical evidence found in Figure 4, we use an 11-day event study. Table 13 reports the results from the analysis. Across the abnormal trading metrics, the coefficients on day 0 is positive and significant. Column (1) reports abnormal total volume and column (2) reports abnormal total trades. Abnormal total volume increased by 1.53 million on event day relative to the estimation period and abnormal total trades increased by 9,324 on event

[11] Abnormal lit trades and abnormal hidden trades show similar patterns to abnormal lit volume (Panel A) and abnormal hidden volume (Panel B), respectively, and are not reported.

day relative to the estimation period. We separate total volume and trades into its lit and hidden portions to identify where the liquidity is stemming from. Column (3) reports abnormal lit volume and column (4) reports abnormal lit trades. Column (5) reports abnormal hidden volume and column (6) reports abnormal hidden trades. Abnormal lit volume increased by 1.29 million on event day relative to the estimation period and abnormal lit trades increased by 6,860 on event day relative to the estimation period. Abnormal hidden volume increased by 234,000, and abnormal hidden trades increased by 2,464 on event day relative to the estimation period. As another measure of trading activity, we examine canceled trades. Column (7) reports the abnormal canceled trades around the event window. Abnormal canceled trades reached 66,075 on day 0. The increased level of trading activity extended after day 0 with some metrics staying above average through day 4. Overall, the results from the event study are consistent with the graphical evidence.

To further test the results in Figure 4, we use an OLS regression controlling for firm and bond characteristics. Regression results are reported in Table 14. All trading metrics are positive and significant on event day with increased trading up to two days after the event. Column (1) reports the results for abnormal total volume and column (2) reports the results for abnormal total trades. Abnormal total volume increased by 1.54 million and abnormal total trades increased by 7,752 on event day after accounting for control variables. Columns (3) and (4) report the results for abnormal lit volume and abnormal lit trades, respectively. Abnormal lit volume increased by 1.29 million and abnormal lit trades increased by 5,655 on event day after accounting for control variables. Columns (5) and (6) report the results for abnormal hidden volume and abnormal hidden trades, respectively. Abnormal hidden volume increased by 239,000, and abnormal hidden trades increased by 2,096 on event day after accounting for control variables. Most of the increase in trading activity is driven by lit trading activity which lasts up to two days post event, while hidden trading activity only increases on event day. Column (7) report the regression results for abnormal canceled trades. Abnormal canceled trades increased by 45,267 on event day after accounting for control variables. Overall, all abnormal trading metrics are positive and significant on day 0 supporting hypothesis 3 and hypothesis 4 in that equity trading increases around the day the convertible bonds go ITM.

5. Conclusion

While a sizeable part of the corporate bond market is made up of convertible bonds, research on the trading of convertible bonds is scarce. Past literature mainly focuses on convertible bond issuances, design decisions, convertible arbitrage, and convertible calls. We explore how convertible bonds trade when the option to convert is ITM. Changes to the option value of convertibles provide a unique setting to isolate the motivation behind trading in the illiquid corporate bond market. Empirical results show that trading activity in convertible bonds increases when the option is ITM. In a market that is illiquid with relatively higher transaction costs, the increased trading activity in ITM convertibles shows that investor demand increases around the time when the conversion option is valuable.

When convertible bonds are ITM, this indicates that the stock price of the firm has exceeded the conversion price which implies an increase in the firm's equity valuation. When the firm is performing well or valued higher by investors, the securities related to the firm will increase in demand. We examine whether ITM convertible bonds lead to a spillover effect on a firm's other bonds. The results from this analysis indicate that trading activity increased in non-convertible bonds of firms with ITM convertible bonds, which confirms a spillover effect.

Since the conversion price is dependent on the stock price of the firm, we expect increased trading in the equity market when the convertible bond becomes convertible. We explore the equity market's reactions to when convertible bonds are ITM and find increases in trading activity and hidden liquidity. Total volume and total trades (comprising of lit and hidden volume and trades) and canceled trades all increased on the day the convertible bonds go in-the-money.

In conclusion, we use the optionality feature of convertible bonds to examine changes in trading behavior when the conversion option is ITM. Trading activity increased in the convertibles with spillover effects to the firms' other bonds and trading activity increased in the equity markets. Our results offer new perspectives in understanding the opaque and illiquid corporate bond market. Future work in convertible bonds can examine whether and when these bonds are converted, whether conversion signals positive future growth prospects of the firm (related to equity returns), whether convertible bond issuers are serial issuers of convertibles, what happens to the trading of OTM convertibles after they were close to being ITM, among others.

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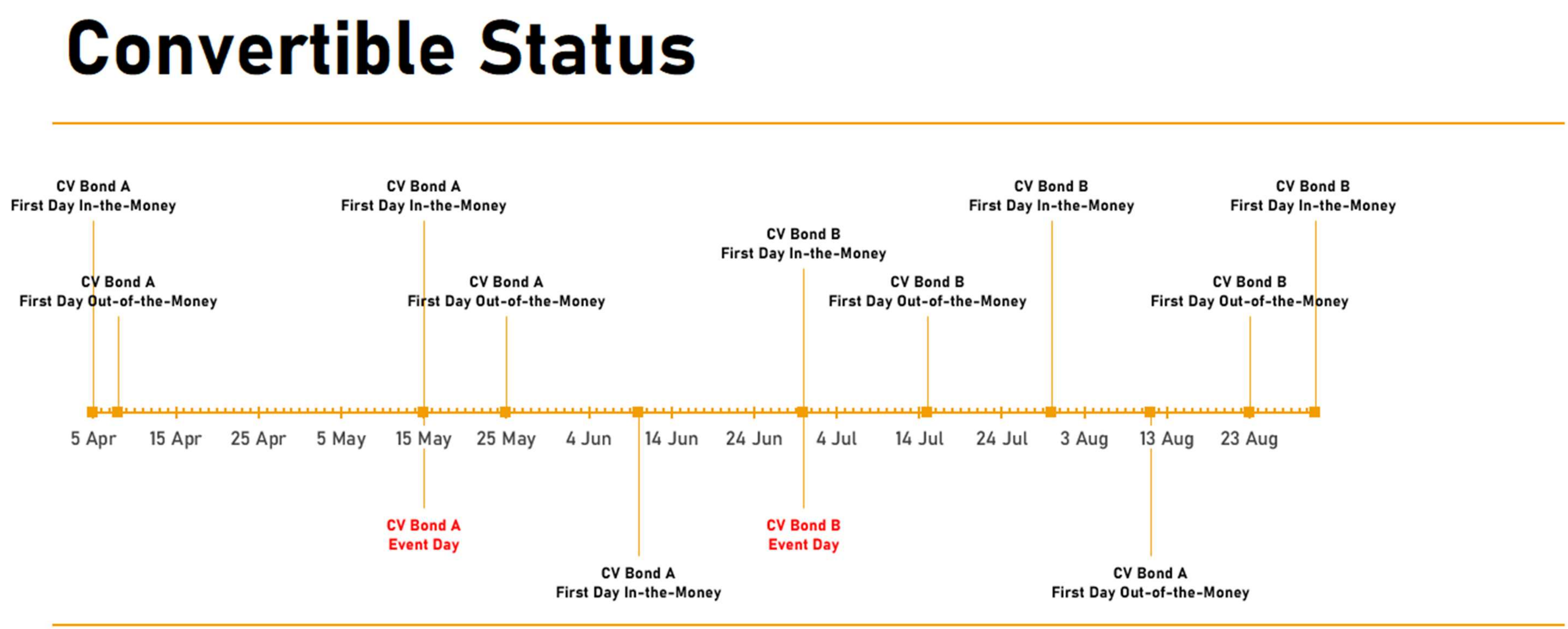
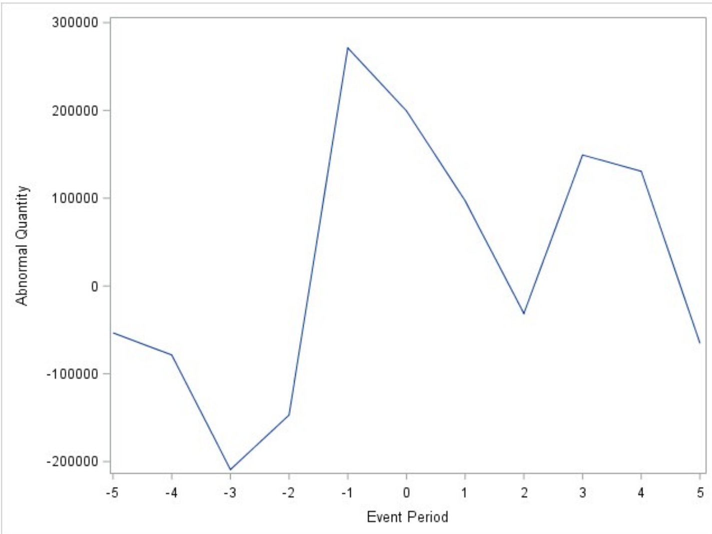


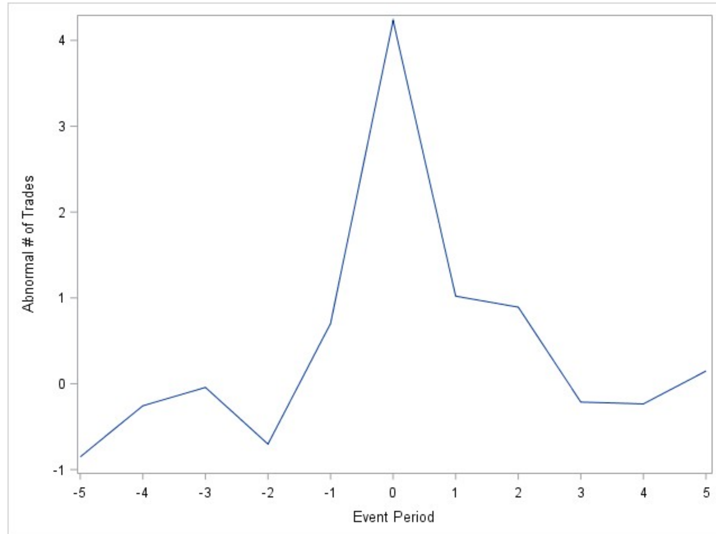
Figure 1. Timeline of Convertible Status

This figure depicts what event window is used when convertible bonds go in- and out-of-the-money throughout the sample period. CV Bond A is a hypothetical convertible bond named A and CV Bond B is a hypothetical convertible bond named B. Event Day denotes the first day the convertible bond goes in-the-money and stays in-the-money for at least 5 days.

Panel A: Abnormal Quantity



Panel B: Abnormal Number of Trades



Panel C: Abnormal Volume

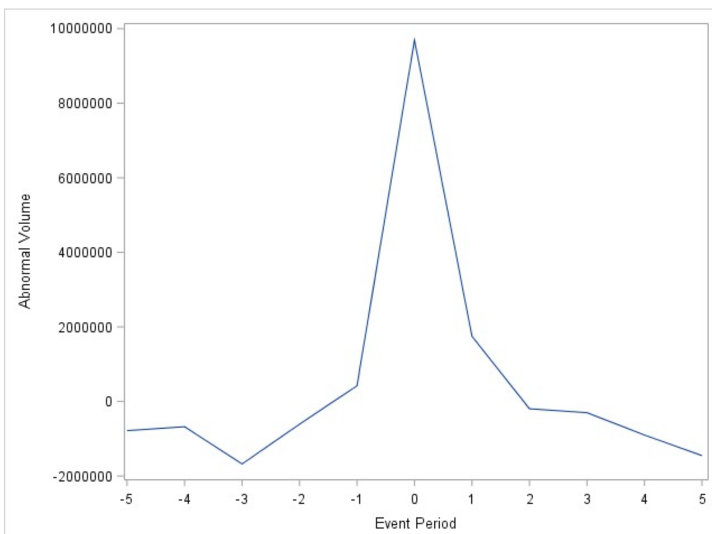
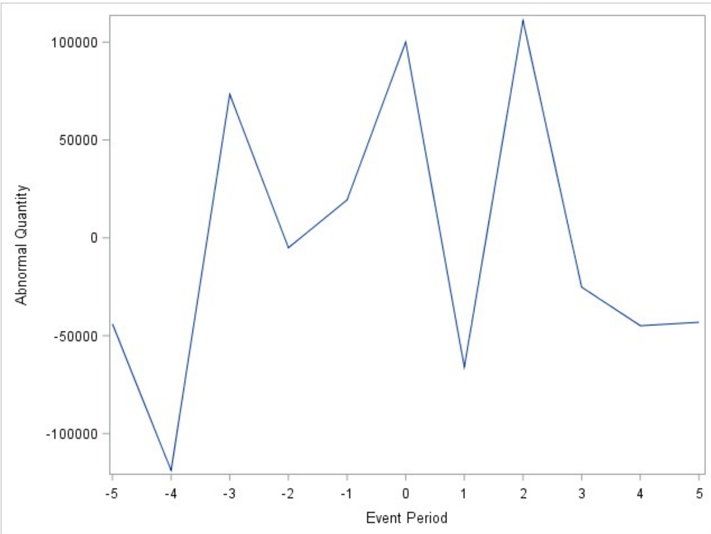


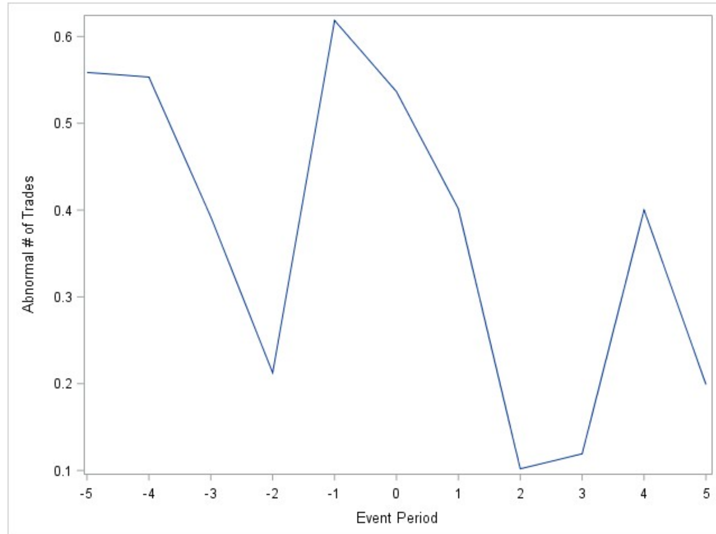
Figure 2. Trading measures of in-the-money convertible bonds.

This figure graphs the abnormal quantity traded (Panel A), abnormal number of trades (Panel B), and abnormal trading volume (Panel C) of convertible bonds that went in-the-money between the event window of [-5, +5].

Panel A: Abnormal Quantity



Panel B: Abnormal Number of Trades



Panel C: Abnormal Volume

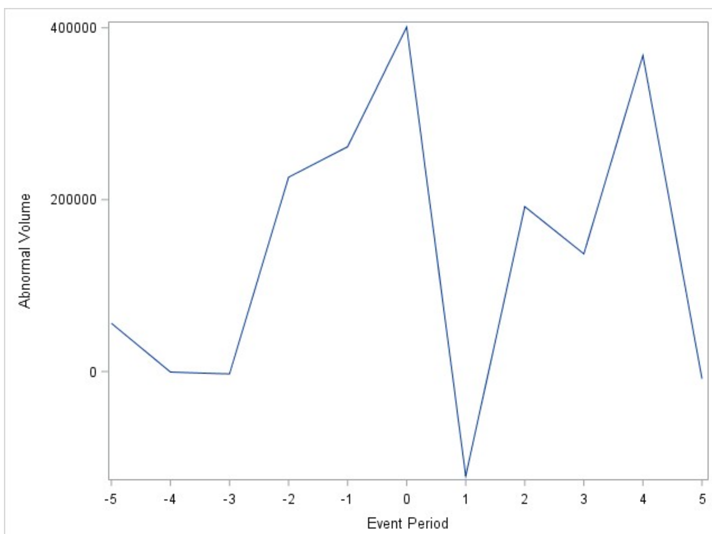
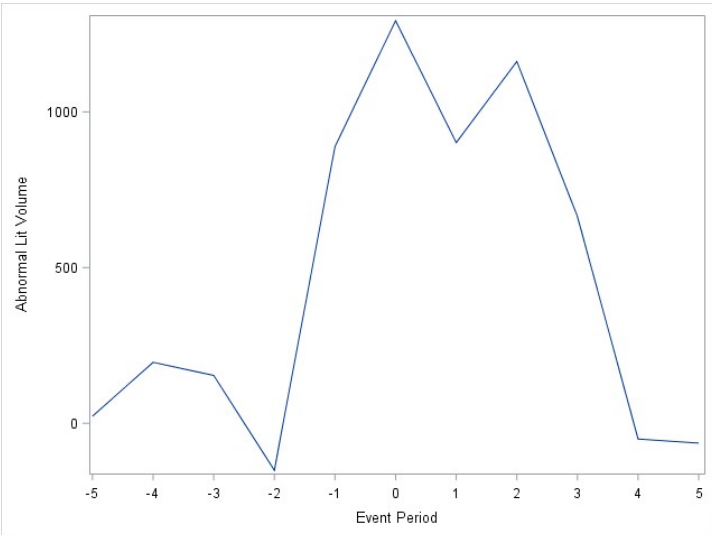


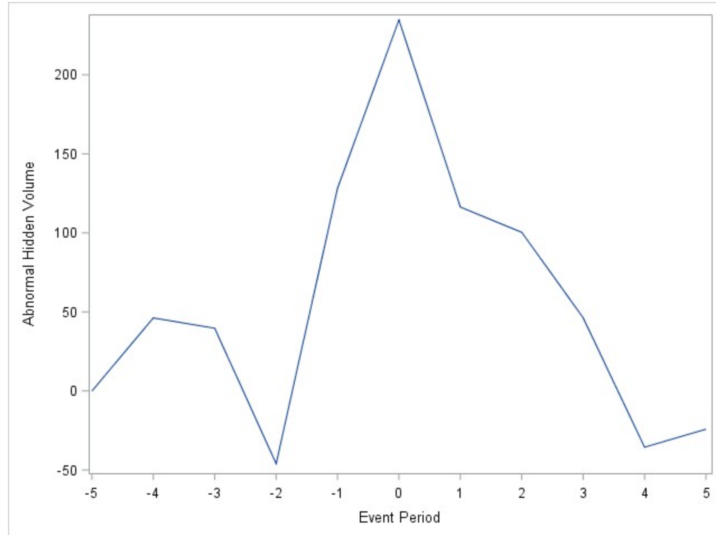
Figure 3. Trading measures of non-convertible bonds.

This figure graphs the abnormal quantity traded (Panel A), abnormal number of trades (Panel B), and abnormal trading volume (Panel C) of non-convertible bonds of firms with in-the-money convertible bonds (spillover effects) during the event window of [-5, +5].

Panel A: Abnormal Lit Volume



Panel B: Abnormal Hidden Volume



Panel C: Abnormal Canceled Trades

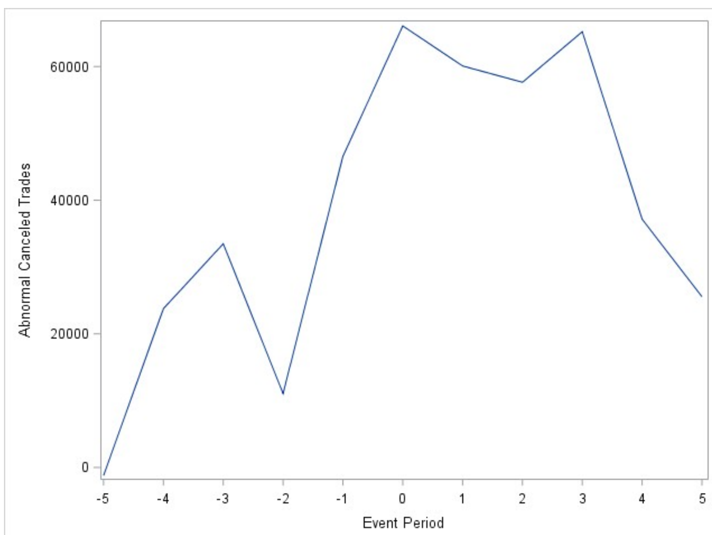


Figure 4. Trading measures of equity market.

This figure graphs the abnormal lit volume (Panel A), abnormal hidden volume (Panel B), and abnormal canceled trades (Panel C) of the equity trading of firms with convertible bonds that went in-the-money during the event window of [-5, +5].

Table 1
Sample Selection
This table reports the sample selection process.

	No. of Bonds
After Standard TRACE Filters	4,163
Sample from Bloomberg after Filters	582
Matched in TRACE and Bloomberg	298
Less Than 10 Trades	7
Not In-the-Money for 5 Days or Not Get In-the-Money	148
No Full Estimation Window	96
Total Bonds	47

Table 2
Descriptive Statistics
This table reports the descriptive statistics for convertible bonds that became in-the-money between January 2015 to December 2022 over the [-5, +5] event window. Number of trades is the daily average number of trades. Trade size is the average daily quantity traded. Trade volume is the product of the daily average number of trades and the average daily trade size. Years to maturity is the difference between the event year and the maturity year. The issue amount is the bond issuance amount in thousands. Price is the issuing firm's closing stock price from CRSP. Sales is the annual sales amount (in millions) sourced from Compustat. Firm size is calculated from daily closing price and shares outstanding from CRSP. ROA is defined as operating income before depreciation over total assets. Leverage is the sum of total debt in current liabilities and total long-term debt divided by stockholders' equity.

Variable	Mean	Std. Dev.	Min	Max
Number of Trades	4.93	5.92	0.27	23.00
Volume	5,290,898	7,134,021	292,909	32,540,545
Trade Size	804,378	494,346	136,909	2,080,197
Years to Maturity	4.80	1.41	3.00	10.00
Issue Amount ('000)	412,347	244,926	115,500	1,150,000
Stock Price	63.91	63.62	4.73	289.47
Sale	2,286.78	3,573.00	36.91	17,337.00
Firm Size	6,310,700	7,956,083	826,879	39,210,799
ROA	-0.07	0.19	-0.53	0.47
Leverage	0.50	3.53	-12.80	10.62
Observations	41			

Table 3
Number of Times a Convertible Bond went In-the-Money
This table reports the number of times a convertible bond went in-the-money during the sample period of January 2015 to December 2022.

Frequency	No. of Bonds
1	5
2	4
3	3
4	3
5	3
6	4
7	8
8	5
9	1
11	2
12	1
13	2
14	1
15	1
19	1
21	1
23	2
Total	47

Table 4
Time of Occurrence
This table reports when the convertible bonds went in-the-money during the sample period of January 2015 to December 2022. Panel A reports the number of convertible bonds that went in-the-money in each month. Panel B reports the number of convertible bonds went that in-the-money in each year.

Panel A: Month of Occurrence	No. of Bonds
January	4
February	1
March	2
April	4
May	1
June	2
July	8
August	3
September	2
October	6
November	9
December	5
Total	47
Panel B: Year of Occurrence	
2015	0
2016	1
2017	0
2018	2
2019	5
2020	19
2021	13
2022	7
Total	47

Table 5
Types of Information Event
This table reports the types of information event that resulted in a convertible bond to be in-the-money. Earnings Announcements include news related to quarterly earnings announcements, earnings estimates from the firm or analysts, or news regarding when the firm is set to announce earnings. Macroeconomic news include news related to how the overall economy is related to the firm's operations. Firm-specific news include news related to mergers and acquisitions, partnerships, patent issuances, government contracts, CEO turnover, new product releases, etc. Analyst forecasts include news regarding analyst upgrades of stock price targets, earnings targets, EPS targets, or future forecasts. Unknown include firms with no tangible news that might affect stock price.

	No. of Bonds
Earnings Announcements	7
Macroeconomic News	1
Firm-Specific News	20
Analyst Forecasts	18
Unknown	1
Total	47

Table 6
Event Study of Trading Metrics
This table presents results from an 11-day event study around in-the-money convertible bonds. Column (1) reports the results for abnormal trading quantity. Column (2) reports the results for abnormal number of trades. Column (3) reports the results for abnormal trading volume. The abnormal measures are computed by taking the daily measures minus the estimation window daily average, where the estimation window is measured from the preceding 25 trading days [-30, -6]. The *t*-tests test whether the abnormal trading measures are significantly different from zero. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

	Abnormal Quantity (1)	Abnormal Trades (2)	Abnormal Volume (3)
t-5	-53,370.75	-0.85	-782,045.96
t-4	-78,552.08	-0.26	-676,897.02
t-3	-209,085.35	-0.04	-1,677,088.51*
t-2	-146,871.12	-0.70	-609,662.98
t-1	271,171.61	0.70	420,464.68
Day 0	199,421.13	4.23***	9,693,890.21**
t+1	96,976.25	1.02	1,746,911.49
t+2	-31,300.32	0.89	-196,705.53
t+3	149,227.28	-0.21	-301,024.68
t+4	130,703.30	-0.23	-900,918.30
t+5	-65,193.76	0.15	-1,452,024.68

Table 7
OLS Regression of Trading Metrics
This table reports the regression results. The dependent variables are the abnormal trading measures. Seven dummy variables are included to capture the seven days around the event day. Other independent variables include the log of bond issue amount, log of annual sales, log of firm size, ROA, leverage, years to maturity, and a buy/sell indicator (1+side log-transformed) representing the side of a trade where side is +1 for buys, -1 for sells, and 0 for no trade. The t-statistics are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

Variables	(1) Abnormal Quantity	(2) Abnormal Trades	(3) Abnormal Volume
$Event_{t-3}$	-219,518.21	0.29	-770,946.79
	(-1.07)	(0.31)	(-0.42)
$Event_{t-2}$	-121,657.11	-0.31	509,811.47
	(-0.59)	(-0.33)	(0.27)
$Event_{t-1}$	321,243.41	1.20	1,711,861.18
	(1.57)	(1.30)	(0.93)
Event Day	177,232.04	4.77***	11,105,857.55***
	(0.88)	(5.21)	(6.10)
$Event_{t+1}$	92,823.48	1.27	2,467,957.28
	(0.47)	(1.41)	(1.37)
$Event_{t+2}$	-40,515.68	1.04	375,325.84
	(-0.20)	(1.13)	(0.20)
$Event_{t+3}$	124,434.91	0.10	503,968.17
	(0.61)	(0.11)	(0.27)
Issue Amount	-311,919.66*	-0.01	-1,449,401.61
	(-1.82)	(-0.01)	(-0.94)
Sale	47,459.99	0.72***	1,643,292.77***
	(0.81)	(2.73)	(3.12)
Firm Size	144,511.18	-0.16	558,269.85
	(1.57)	(-0.38)	(0.67)
ROA	-464,004.55	-2.83**	-4,878,195.73*
	(-1.47)	(-1.97)	(-1.71)
Leverage	18,427.93	-0.01	169,945.71
	(0.90)	(-0.14)	(0.92)
Years to Maturity	-82,786.10*	0.16	-320,803.41
	(-1.84)	(0.79)	(-0.79)
Buy/Sell Side	-136,037.80	-2.33***	-3,872,864.46**
	(-0.74)	(-2.80)	(-2.33)
Constant	1,856,276.66	-3.32	-291,108.42
	(1.35)	(-0.53)	(-0.02)
Observations	480	480	480
R-squared	0.04	0.10	0.12

Table 8
Trading Behavior Pre- and Post-Event
This table compares the trading behavior in the pre-event window of [-5, -1] to the post-event window of [0, 5]. The t-test on the difference between the pre-event and post-event windows examines the total number of buys, sells, customer trades, dealer trades, and agent trades for the in-the-money convertible bonds in the sample. $***$, $**$, and $*$ indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Pre [-5, -1]	Post [0, 5]	Difference	t-stat
Buys	11.40	17.53	-6.13***	-3.97
Sells	9.36	14.62	-5.26***	-3.96
Customer Trade	13.19	21.57	-8.38***	-4.27
Dealer Trade	7.04	9.87	-2.83**	-2.11
Agent Trade	0.40	0.60	-0.20	-1.09

Table 9
Descriptive Statistics – Spillover Effects
This table reports the descriptive statistics for the non-convertible bonds of firms with convertible bonds that became in-the-money between January 2015 to December 2022 over the [-5, +5] event window. Number of trades is the daily average number of trades. Trade size is the average daily quantity traded. Trade volume is the product of the daily average number of trades and the average daily trade size. Years to maturity is the difference between the event year and the maturity year. Price is the issuing firm's closing stock price from CRSP. Sales is the annual sales amount (in millions) sourced from Compustat. Firm size is calculated from daily closing price and shares outstanding from CRSP. ROA is defined as operating income before depreciation over total assets. Leverage is the sum of total debt in current liabilities and total long-term debt divided by stockholders' equity.

Variable	Mean	Std. Dev.	Min	Max
Number of Trades	1.3262	3.6117	0	12.0909
Volume	802,521.51	2,280,952.6	0	8,443,754.5
Quantity	53,756.611	127,210.69	0	384,601.28
Years to Maturity	2.419	2.0114	1	6.6667
Stock Price	54.4119	42.4398	6.3882	125.2945
Sale	2,536.9619	4,036.5826	252.002	17337
Size	6,228,203.3	7,142,077.7	826,878.58	23,842,534
ROA	-0.0508	.2203	-0.4891	0.4689
Leverage	-0.3989	4.7613	-12.8044	10.6245
Observations	18			

Table 10
Event Study – Spillover Effects
This table presents results from an 11-day event study around the non-convertible bonds of firms with in-the-money convertible bonds. Column (1) reports the results for abnormal trading quantity. Column (2) reports the results for abnormal number of trades. Column (3) reports the results for abnormal trading volume. The abnormal measures are computed by taking the daily measures minus the estimation window daily average, where the estimation window is measured from the preceding 25 trading days [-30, -6]. The *t*-tests test whether the abnormal trading measures are significantly different from zero. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

	Abnormal Quantity (1)	Abnormal Trades (2)	Abnormal Volume (3)
t-5	-43,995.97	-0.19	295,241.14
t-4	-118,852.37***	0.30	-606,658.68**
t-3	73,341.25	-0.06	-53,491.52
t-2	-5,053.18	0.28	-174,157.45
t-1	19,350.19	1.30*	426,451.41
Day 0	100,097.39	1.27**	2,036,901.14
t+1	-66,022.48	-0.58**	-398,482.38**
t+2	111,091.90	0.89**	652,125.13
t+3	-25,186.75	-0.15	-336,236.94
t+4	-44,894.14	-0.28	14,401.87
t+5	-43,120.93	0.05	-422,874.92

Table 11
OLS Regression – Spillover Effects
This table reports the regression results testing spillover effects on the trading of non-convertible bonds from firms with in-the-money convertible bonds. The dependent variables are the abnormal trading measures. Seven dummy variables are included to capture the seven days around the event day. Other independent variables include log of annual sales, log of firm size, ROA, leverage, years to maturity, and a buy/sell indicator (1+side log-transformed) representing the side of a trade where side is +1 for buys, -1 for sells, and 0 for no trade. The t-statistics are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

Variables	(1) Abnormal Quantity	(2) Abnormal Trades	(3) Abnormal Volume
$Event_{t-3}$	99,982.88	-0.05	106,874.76
	(1.08)	(-0.08)	(0.13)
$Event_{t-2}$	45,825.49	0.38	-30,980.84
	(0.50)	(0.59)	(-0.04)
$Event_{t-1}$	70,399.02	1.31**	568,820.48
	(0.77)	(2.02)	(0.72)
Event Day	166,319.16*	1.27*	2,225,700.82***
	(1.82)	(1.95)	(2.82)
$Event_{t+1}$	-1,842.34	-0.60	-297,437.28
	(-0.02)	(-0.92)	(-0.37)
$Event_{t+2}$	130,057.52	0.85	672,739.06
	(1.42)	(1.30)	(0.85)
$Event_{t+3}$	30,796.24	-0.18	-234,792.44
	(0.34)	(-0.27)	(-0.30)
Sale	-23,954.07	-0.08	-185,911.27
	(-0.79)	(-0.38)	(-0.71)
Firm Size	24,057.47	0.16	566,698.57
	(0.56)	(0.52)	(1.54)
ROA	-162,211.84	-2.71*	-3,480,564.20*
	(-0.73)	(-1.72)	(-1.81)
Leverage	2,991.73	0.03	177,689.24**
	(0.36)	(0.49)	(2.47)
Years to Maturity	6,837.23	0.04	92,649.97
	(0.90)	(0.67)	(1.41)
Buy/Sell Side	-231,489.76	-1.11	-2,002,831.98
	(-1.46)	(-0.99)	(-1.46)
Constant	-261,154.39	-2.08	-7,476,310.62
	(-0.45)	(-0.50)	(-1.48)
Observations	789	789	789
R-squared	0.0118	0.0200	0.0312

Table 12
Descriptive Statistics – Equity Market
This table reports the descriptive statistics in the equity market for firms with convertible bonds that became in-the-money between January 2015 to December 2022 over the [-5, +5] event window. Price is the closing price from CRSP. Sales is the annual sales amount (in millions) sourced from Compustat. Firm size is calculated from daily closing price and shares outstanding from CRSP. ROA is defined as operating income before depreciation over total assets. Leverage is the sum of total debt in current liabilities and total long-term debt divided by stockholders' equity. Total Volume ('000) and Total Trades are calculated as the sum of lit and hidden volume and lit and hidden trades, respectively, from MIDAS.

Variable	Mean	Std. Dev.	Min	Max
Price	63.92	63.62	4.73	289.47
Sale (millions)	2,286.78	3,573.00	36.91	17,337.00
Firm Size	6,310.95	7,955.94	826.88	39,210.80
ROA	-0.07	0.19	-0.53	0.47
Leverage	0.50	3.53	-12.80	10.62
Total Volume ('000)	1,956.8194	5,641.8529	18.9544	32,874.451
Total Trades	12,309.87	17,581.798	1,368.88	105,907.04
Canceled Trades	142,530.18	186,476.47	20,959.68	1,122,612.00
Observations	41			

Table 13
Event Study – Equity Market
This table presents results from an 11-day event study on equity trading of firms with convertible bonds that went in-the-money during the sample period. Column (1) reports the results for abnormal total volume (in ‘000). Column (2) reports the results for abnormal total trades. Column (3) reports the results for abnormal lit volume (in ‘000). Column (4) reports the results for abnormal lit trades. Column (5) reports the results for abnormal hidden volume (in ‘000). Column (6) reports the results for abnormal hidden trades. Column (7) reports the results for abnormal canceled trades. The abnormal measures are computed by taking the daily measures minus the estimation window daily average, where the estimation window is measured from the preceding 25 trading days [-30, -6]. The *t*-tests test whether the abnormal trading measures are significantly different from zero. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

	Abnormal Total Volume (1)	Abnormal Total Trades (2)	Abnormal Lit Volume (3)	Abnormal Lit Trades (4)	Abnormal Hidden Volume (5)	Abnormal Hidden Trades (6)	Abnormal Canceled Trades (7)
t-5	22.77	-60.20	22.85	12.47	-0.08	-72.67	-1,201.19
t-4	241.92*	3,591.69*	195.73**	2,538.93**	46.19	1,052.76	23,762.48*
t-3	193.29	3,679.17**	153.64	2,690.34**	39.66	988.83	33,483.63***
t-2	-196.77	944.30	-150.81	861.71	-45.96	82.59	11,056.96
t-1	1,016.42	4,652.19	888.19	3,730.34	128.23	921.85	46,579.05
Day 0	1,527.51*	9,324.30***	1,292.78*	6,860.19***	234.73**	2,464.11***	66,075.46***
t+1	1,017.42*	5,801.84**	901.09*	4,604.27*	116.33**	1,197.56**	60,082.31*
t+2	1,262.15	5,849.08**	1,161.82	4,833.91*	100.33	1,015.17**	57,645.05**
t+3	709.84*	4,810.45*	663.56*	4,126.82*	46.28	683.63*	65,212.68*
t+4	-86.19	1,540.40	-50.60	1,290.14*	-35.59	250.26	37,143.66**
t+5	-87.79	1,204.99	-63.60	1,053.77	-24.19	151.22	25,553.46

Table 14
OLS Regression – Equity Market
This table presents the regression results in the equity market for firms with convertible bonds that went in-the-money during the sample period. Column (1) reports the results for abnormal total volume. Column (2) reports the results for abnormal total trades. Column (3) reports the results for abnormal lit volume. Column (4) reports the results for abnormal lit trades. Column (5) reports the results for abnormal hidden volume. Column (6) reports the results for abnormal hidden trades. Column (7) reports the results for abnormal canceled trades. The abnormal measures are computed by taking the daily measures minus the estimation window daily average, where the estimation window is measured from the preceding 25 trading days [-30, -6]. The t-statistics are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% levels, respectively.

	Abnormal Total Volume (1)	Abnormal Total Trades (2)	Abnormal Lit Volume (3)	Abnormal Lit Trades (4)	Abnormal Hidden Volume (5)	Abnormal Hidden Trades (6)	Abnormal Canceled Trades (7)
$Event_{t-3}$	166.75	2,175.08	122.48	1,486.07	44.27	689.02	12,131.09
	(0.29)	(0.92)	(0.24)	(0.78)	(0.60)	(1.17)	(0.50)
$Event_{t-2}$	-224.20	-617.08	-180.81	-358.75	-43.39	-258.33	-10,436.31
	(-0.38)	(-0.26)	(-0.35)	(-0.19)	(-0.59)	(-0.44)	(-0.43)
$Event_{t-1}$	1,019.05*	3,159.97	884.70*	2,574.81	134.35*	585.16	25,938.96
	(1.75)	(1.33)	(1.70)	(1.36)	(1.83)	(0.99)	(1.06)
Event Day	1,534.12***	7,752.46***	1,294.21**	5,655.93***	239.91***	2,096.53***	45,267.67*
	(2.63)	(3.26)	(2.49)	(2.98)	(3.26)	(3.56)	(1.85)
$Event_{t+1}$	1,018.65*	4,194.13*	899.27*	3,396.85*	119.38	797.29	39,540.62
	(1.74)	(1.77)	(1.73)	(1.79)	(1.62)	(1.35)	(1.62)
$Event_{t+2}$	1,268.53**	4,279.56*	1,165.39**	3,655.50*	103.14	624.07	37,086.83
	(2.17)	(1.80)	(2.24)	(1.92)	(1.40)	(1.06)	(1.52)
$Event_{t+3}$	710.95	3,214.66	663.00	2,937.92	47.95	276.74	45,088.26*
	(1.22)	(1.35)	(1.27)	(1.55)	(0.65)	(0.47)	(1.84)
Price	-6.98**	0.40	-6.98**	-10.22	-0.00	10.62***	-145.67
	(-2.24)	(0.03)	(-2.52)	(-1.01)	(-0.01)	(3.38)	(-1.12)
Sale	340.09**	3,032.01***	321.91**	2,593.89***	18.18	438.13***	39,842.62***
	(2.03)	(4.45)	(2.16)	(4.77)	(0.86)	(2.60)	(5.69)
Firm Size	287.61	1,185.26	256.52	935.31	31.10	249.94	6,319.43
	(1.27)	(1.28)	(1.27)	(1.27)	(1.09)	(1.09)	(0.66)
ROA	-1,140.86	-13,630.65***	-969.06	-10,794.03***	-171.79	-2,836.62***	-121,102.75***
	(-1.24)	(-3.64)	(-1.18)	(-3.60)	(-1.48)	(-3.05)	(-3.14)
Leverage	48.05	-93.81	42.09	-96.06	5.97	2.24	-1,841.19
	(0.93)	(-0.45)	(0.91)	(-0.57)	(0.92)	(0.04)	(-0.85)
Constant	-6,324.21**	-38,405.57***	-5,705.96**	-30,971.83***	-618.25	-7,433.74**	-347,527.63***
	(-2.10)	(-3.14)	(-2.13)	(-3.16)	(-1.63)	(-2.45)	(-2.76)
Observations	495	495	495	495	495	495	495
R-squared	0.06	0.11	0.06	0.11	0.04	0.11	0.12